

ARCHIMEDES

Preparing DICOM Data for Submission

A guide for project uploading imaging(DICOM) data to ARCHIMEDES

ARCHIMEDES accepts DICOM imaging data uploaded in batches to a shared drive or SFTP location. Before your data can be ingested we need to know which participant and which visit each scan belongs to.

You only need to provide two things

A folder tree organised by participant and visit, and two small tab-separated files: participants.tsv and a sessions.tsv per participant. You do not create any records in ARCHIMEDES; participant creation, identifiers and archiving to the platform are handled on our side.

1. Folder structure

Your project has a dedicated upload location. Inside it, use one folder per participant, and inside that one folder per visit. Place the DICOM files for that visit inside the visit folder.

```
<project_name>/deidentified-raw/imaging/dicoms/
├── participants.tsv
├── sub-ARCHI0001/
│   ├── sub-ARCHI0001_sessions.tsv
│   ├── ses-V01/
│   │   ├── MR/      (MRI DICOM files)
│   │   └── PET/      (PET DICOM files)
│   └── ses-V02/
│       └── MR/
└── sub-ARCHI0002/
    ├── sub-ARCHI0002_sessions.tsv
    ├── ses-V01/
    └── MR/
```

Only the top two levels are fixed: the sub- and ses- prefixes tell us which part of a folder name is your identifier. Below the visit folder, arrange things however suits you; separating by modality, as above, is usually the easiest way to keep each examination apart, but any layout works.

2. Naming rules

Identifiers are yours to choose. Keep them to letters and numbers; underscores and dashes inside the identifier cause problems downstream. The sub- and ses- prefixes are separate and always required as it will help us to identify the participants and sessions/visits..

Rule	Good	Avoid
No underscores in identifiers	sub-ARCHI0001	sub-ARCHI_0001, sub-ARCHI-0001
No spaces, slashes or quotation marks	sub-P014	sub-Patient 14
No names or identifying information	sub-UOHI0007	sub-Smith-J
Same identifier for a participant across all their visits	sub-P014/ses-V01 and /ses-V02	sub-P014 then sub-P14
Visit labels short and consistent	ses-V01, ses-Baseline	ses-visit one, ses-V01_visit etc

Visits are not created automatically

Visit labels must already be configured in ARCHIMEDES platform for your project, and a folder using an unconfigured label will be skipped. We set these up with you during onboarding - tell us the labels that fit how your study actually runs and we will configure them before your first upload. If your study adds a visit/session later, let us know and we will add it.

3. The two spreadsheets

Build these in Excel or Google Sheets if that is easiest. What matters is how you save them: choose Tab Delimited Text (Excel) or Tab-separated values (Sheets), not comma-separated and not .xlsx.

participants.tsv - one row per participant, at the top level

IMPORTANT: Location- the top level of your upload, alongside the sub- folders.

Column	Required	Format	Example
participant_id	Yes	Must match the sub- folder name	sub-ARCHI0001
dob	Yes	YYYY-MM-DD	1948-03-01
sex	Yes	Male, Female or Other	Male

```
participant_id  dob      sex
sub-ARCHI0001  1948-03-01  Male
sub-ARCHI0002  1962-11-03  Female
```

De-identification tools commonly replace the date of birth in DICOM headers with a generalised value such as 1 January, and may remove sex entirely, so we need this information from you directly. Send whatever precision you are comfortable with in the following format: YYYY-MM-DD, YYYY-MM, or just YYYY. If you can only provide an age, tell us and we will work out how to handle it.

sessions.tsv — one per participant, inside their sub- folder

IMPORTANT: Location inside that participant's sub- folder, named after it.

For example sub-ARCHI0001_sessions.tsv. One row per visit. Only session_id is required; add any further columns useful to you.

Column	Required	Format	Example
session_id	Yes	Must match the ses- folder name	ses-V01
acq_date	Optional	YYYY-MM-DD, date of the visit	2024-11-19

4. About the DICOM headers

What we read

We use a small number of header fields and ignore the rest.

Field	Why we read it
PatientName	Your identifier for the participant

Field	Why we read it
StudyInstanceUID	Tells us which files belong to the same examination
Modality	Identifies the type of scan — MR, PET, CT — so each examination is archived separately
StudyDate	Used to order examinations within a visit

What we change

On our copy only (the original DICOMs are never changed) PatientName is replaced with the ARCHIMEDES platform identifier for that participant and visit, and OtherPatientName is set to your original PatientName so the link back is preserved. Nothing else in the headers is modified.

If PatientName carries anything you need

We overwrite it, so anything stored there will not survive into ARCHIMEDES. If it holds something meaningful to you — a local code, a protocol label — tell us before uploading and we will agree where it should live instead.

What the headers should not contain

Please de-identify before sending. Headers frequently carry more than people expect:

- Names, initials, medical record numbers or contact details, in any field
- Free-text fields such as StudyDescription, PatientComments and InstitutionName, which often hold notes typed at the scanner
- Private vendor tags, which vary by manufacturer and are easy to overlook

Please keep SeriesDescription

It holds the scan labels. T1w, FLAIR, the PET protocol name and is what makes a study readable in ARCHIMEDES once it is archived. Only modify it if it contains something identifying. Everything else in the list above can be blanked without affecting ingestion.

We assume the data arrives de-identified. If you are unsure what your export contains, send us one study example per vendor/scanner model/scan modality and we will tell you what is in it.

5. What we do with the DICOMs

A single visit folder often contains more than one imaging examination; an MRI and a PET acquired the same day, for instance. You do not need to separate these, or to know anything about StudyInstanceUIDs: we run a script on receipt that splits your upload into one archive per examination before it is loaded.

- Organise by participant and visit; below that, group by modality if it helps you.
- DICOMDIR files and files without extensions are fine - leave them as they are.
- If your export does not fit this shape, talk to us before uploading. There is usually a way to accommodate it.